

NLP Placement Revision Guide

Q: What is NLP?

A: NLP (Natural Language Processing) is a branch of AI that helps computers understand, process, and generate human language.

Q: Why can't ML models directly work on text?

A: Machine learning models work on numerical data, so text must be converted into numerical representations.

Q: What is a Corpus?

A: A corpus is a collection of documents or text data.

Q: What is a Vocabulary?

A: Vocabulary is the set of all unique words present in a corpus.

Q: What is Text Preprocessing?

A: Cleaning and preparing text before modeling, including lowercasing, punctuation removal, stopword handling, stemming, and lemmatization.

Q: What are Stopwords?

A: Frequently occurring words such as 'the', 'is', and 'and' that often carry little useful information.

Q: Why should stopwords not always be removed?

A: Words like 'not' can completely change the meaning of a sentence.

Q: What is Stemming?

A: Reducing words to their root form using heuristic rules. Example: studies → studi.

Q: What is Lemmatization?

A: Reducing words to their meaningful dictionary form using linguistic knowledge. Example: studies → study.

Q: Difference between Stemming and Lemmatization?

A: Stemming is faster but less accurate. Lemmatization is more meaningful and accurate.

Q: What is Tokenization?

A: The process of splitting text into smaller units called tokens.

Q: Types of Tokenization?

A: Word Tokenization and Sentence Tokenization.

Q: What is Bag of Words (BoW)?

A: A vectorization technique that converts text into numerical vectors by counting word frequencies.

Q: Limitations of BoW?

A: Ignores word order, ignores semantic meaning, and creates large sparse vectors.

Q: What are N-Grams?

A: Sequences of N consecutive words. Examples: Unigram, Bigram, Trigram.

Q: Why are N-Grams useful?

A: They preserve some word order and capture phrases such as 'not good'.

Q: What is TF-IDF?

A: A vectorization technique that gives higher importance to words that are frequent in a document but rare across the corpus.

Q: Why was TF-IDF introduced?

A: BoW gives equal importance to all words.

Q: What is Text Classification?

A: Assigning predefined labels to text, such as Positive/Negative or Spam/Ham.

Q: Examples of Text Classification?

A: Sentiment Analysis, Spam Detection, News Classification.

Q: What is Accuracy?

A: Correct predictions divided by total predictions.

Q: What is a Confusion Matrix?

A: A table showing TP, TN, FP, and FN.

Q: What is Precision?

A: $TP / (TP + FP)$.

Q: What is Recall?

A: $TP / (TP + FN)$.

Q: What is F1 Score?

A: The harmonic mean of Precision and Recall.

Q: What is NER?

A: Named Entity Recognition identifies entities such as Person, Organization, Location, and Date.

Q: Examples of NER labels?

A: PERSON, ORG, GPE, DATE.

Q: Difference between NER and POS Tagging?

A: NER identifies entities; POS Tagging identifies grammatical roles.

Q: What is POS Tagging?

A: Assigning grammatical categories such as NOUN, VERB, and ADJECTIVE to words.

Q: What is Word2Vec?

A: A word embedding technique that creates dense vectors where words appearing in similar contexts have similar vectors.

Q: Why was Word2Vec introduced?

A: TF-IDF cannot capture semantic relationships between words.

Q: What are Dense and Sparse Vectors?

A: BoW/TF-IDF produce sparse vectors; Word2Vec produces dense vectors.

Q: What is a Sentence Embedding?

A: A single vector representation of an entire sentence.

Q: Limitation of Word2Vec?

A: A word always gets the same vector regardless of context.

Q: What is Polysemy?

A: A word having multiple meanings, such as 'bank'.

Q: Why was BERT introduced?

A: To generate contextual embeddings where the same word can have different vectors in different contexts.

Q: Difference between Word2Vec and BERT?

A: Word2Vec creates fixed embeddings; BERT creates contextual embeddings.

Q: What is Cosine Similarity?

A: A measure of similarity between vectors based on the angle between them.

Q: Why use Cosine Similarity in NLP?

A: It compares semantic similarity between embeddings.

Q: What is Topic Modeling?

A: An unsupervised technique used to discover hidden topics in documents.

Q: What is LDA?

A: Latent Dirichlet Allocation, a popular topic modeling algorithm.

Q: Difference between Text Classification and Topic Modeling?

A: Classification uses predefined labels; Topic Modeling discovers topics automatically.

Q: What is Naive Bayes?

A: A probabilistic classification algorithm based on Bayes Theorem.

Q: Why is it called Naive?

A: It assumes all features are independent of each other.

Q: Why is Naive Bayes popular in NLP?

A: It is simple, fast, and works well on high-dimensional text data.

Q: Difference between Logistic Regression and Naive Bayes?

A: Naive Bayes is faster; Logistic Regression often achieves better accuracy with enough data.

Q: Explain NLP Evolution.

A: BoW → TF-IDF → Word2Vec → BERT. Each technique solved limitations of the previous one.